Developing an Oxygen-Sensitive Biological Gene Circuit to Increase Hydrogen Production in *Synechocystis* PCC 6803 *in silico*

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Fall 2019

Introduction

As fossil fuel reserves deplete and greenhouse gases warm our atmosphere, the world is looking for a cheap, sustainable alternative energy source. One possible answer is manipulating biological organisms to create biomass that can then be converted into energy. In this field, cyanobacteria are of great interest because of their malleable genome and innate autotrophic cellular machinery. There already exists technology capable of producing energy from hydrogen; thus, if scientists can engineer organisms to produce mass amounts of biomass, the world will be supplied with energy.

Previous research has demonstrated how cyanobacteria can produce hydrogen under specific conditions, such as removing sulfur from algae (Melis et al., 2000). Current research focuses on how to optimize the innate hydrogen production pathways that cyanobacteria possess, but none have rendered a commercially viable formula. This paper presents a novel genetic circuit to increase the hydrogen production of *Synechocystis* PCC 6803, a well-studied photosynthetic bacterium. In this organism, an enzyme named hydrogenase catalyzes the reaction $2H^{+} + 2e^{-} \leftrightarrow 2H_2$. This enzyme functions as a bidirectional hydrogenase, reducing protons to dihydrogen in addition to oxidizing apart H_2. It has a nickel-iron core, and reduces NAD(P)H in order to gain electrons (Appel et al., 1996). During photosynthesis, cyanobacteria store high energy electrons in a "PQ Pool," or the plastoquinone pool, located in the thylakoid membrane. Researchers believe that hydrogenase takes the lower-energy electrons from here—the ones that do not have enough energy to invoke the next step of photosynthesis—and uses them to create hydrogen, therefore not wasting any energy (Appel et al., 2000). More recently, they have also found that when photosynthesis is down-regulated—for example, during nighttime—hydrogen production by hydrogenase is heavily favored (De Rosa et al., 2015). However, hydrogenase is greatly inhibited by oxygen. There have been attempts to artificially create a semi-permeable membrane using nanotechnology that blocks oxygen or to genetically reengineer the

hydrogenase to be tolerant to oxygen, but neither of these have produced hydrogen at a significant level (Santos et. al; Xu et al., 2005).

No previous attempts have been made to create a dynamic genetic circuit that facilitates hydrogen production by responding to the oxygen level. The following paper details a novel biological circuit and shows that it prevents oxygen from building up in the cell, thereby aiding hydrogen production. It also suggests future steps in order to prove this circuit works *in vitro*.

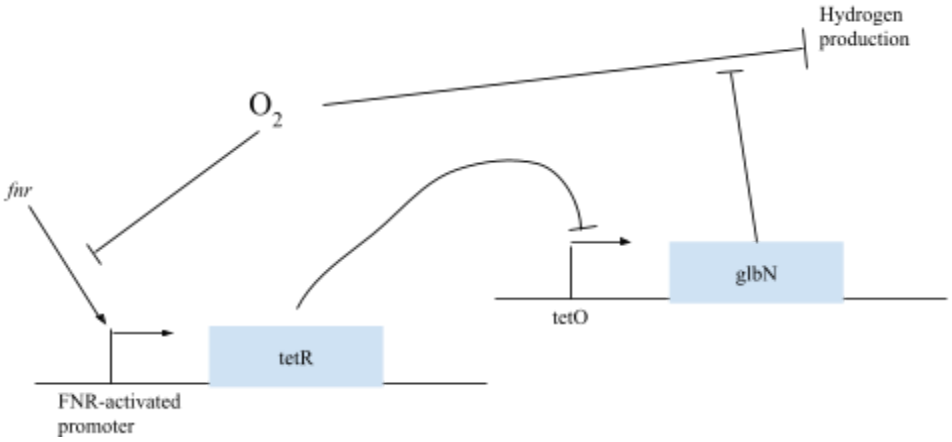

Figure 1: Pictorial representation of the circuit. If oxygen is present, then tetR cannot be transcribed, allowing glbN to be produced and bind to O_2. If oxygen is not present, then fnr *is in its active state, and tetR represses the production of glbN.*

Methods

Oxygen can diffuse through the membrane and be present in the cytoplasm of bacteria. In order to detect oxygen, this circuit will integrate *Escherichia coli's* fumarate and nitrate reduction (FNR) system into my organism. In *E. coli*, this system is responsible for detecting anaerobic conditions and regulating gene expression in response. There have already been successful attempts to create an oxygen-dependent synthetic circuit in *Synechocystis* PCC 6803 using this system (Immuthen et al., 2016). The transcription factor will be *fnr* from *E. Coli MG1655* (Gene ID: 945908). This protein will be constitutively expressed by promoter Bba J23104. The *fnr* protein is present in the cytoplasm of the cell, and when it interacts with oxygen, it transforms from its active state to its inactive state. A $[4Fe\text{-}4S]^{2+}$ cluster transforms into a $[2Fe\text{-}2S]^{2+}$ cluster in the presence of oxygen (Khoroshilova et al., 1997). The $[2Fe\text{-}2S]^{2+}$ cluster is unable to bind to DNA, thereby deactivating *fnr*. This reaction occurs only when O_2 is present, but O_2 is not used in the reaction (Unden et al., 1997). The activated *fnr* complex will be referred to as *fnr*–a in this paper, and the deactivated complex will be referred to as *fnr*–d. The deactivated complex, which can either be the $[2Fe\text{-}2S]^{2+}$ cluster or the apoFNR form that occurs with prolonged exposure to O_2, can transform back into the activated form of the $[4Fe\text{-}4S]^{2+}$ cluster in the presence of glutathione (Tran et al., 2000). This paper assumes that there will always be enough glutathione

present for this reaction to occur. The corresponding FNR-activated promoter will be used to control gene expression, hereby referred to as fpromoter (Grainger et al., 2007).

This promoter will be upstream of the commonly used TetR repressor. This circuit will specifically use tetR because it has been shown to work in bacteria before (Vavitsas et al., 2019). It is assumed that the off-target effects of tetR are minimal and neglectable and that there is an abundance of RNA polymerase. It is also assumed that all mRNA dynamics are at steady state because mRNA dynamics are much faster and reach steady state faster than protein dynamics. tetR has a nucleotide length of 207, and if RNAP operates at 40-80 nt/sec, then it should take between 2.6-5.2 seconds for one to be transcribed. This is equivalent to 11.5-23 per minute.

Once tetR is transcribed, it will be able to bind to tetO. tetO will be the promoter of a globin protein. Specifically, it will be transcribing glbN, a protein already present in *Synechocystis* PCC 6803 which cooperatively binds to oxygen (Thorsteinsson et al., 1999). There will be one globin protein transcribed every 1.55-3.1 seconds, which is 19.4-38.7 per minute. It binds with cooperativity P_{50} = 0.013 mmHg at 20°C (Couture et al., 1999). Furthermore, glbN resides on the peripheral membrane of the cell, so ideally it will bind to oxygen before oxygen even enters the cell, allowing the cell to increase its hydrogen production (Hill et al., 1996). There is also the need to consider the oxygen produced by the cell in photosynthesis. Since this oxygen will already be inside the cell, it is assumed that glbN will bind to it when it is in the cytoplasm. When oxygen is present, glbN will be expressed because tetR will not be repressing tetO, but when the conditions are anaerobic, glbN will not be produced. The reason for this is to limit the load placed on the cell so that the cell is only producing glbN when it needs to. Otherwise, when oxygen is not present, the cell can focus its resources on other processes.

The reason for the end product to be a globin protein instead of something else, such as hydrogenase, is that if oxygen is present it will inhibit hydrogenase, no matter the quantity of hydrogenase. Thus, the rationale for producing globin is so the cell could still hypothetically create hydrogen, even when producing oxygen with photosynthesis. In addition, hydrogen production in *Synechocystis* only happens in specific conditions, one of which is immediately when the cell starts to photosynthesize. Thus, designing a circuit that produces hydrogenase would not be effective because of the time delay involved in the circuit. Hydrogenase needs to be present and uninhibited in order for maximal hydrogen production. Once O_2 is no longer inhibiting hydrogenase, the limiting factor of hydrogen production of a cell is the rate at which NAD(P)H is recycled through the NDH-1 complex (Cournac et al., 2002).

The set of ODEs in Figure 2 will be computationally tested using the Biosystem class of Matlab. In order to empirically test this model, a bioreactor would be required to control the inputs of O_2 and measure the output of H_2. First, a baseline must be established, so a colony without any genetic changes in both aerobic/anaerobic environments and dark/light environments would be analyzed. *Synechocystis* PCC 6803 produces hydrogen mainly in two precise environmental states: in the dark, when the cell is forced to assimilate glycogen as an electron

acceptor, and at the beginning of photosynthesis when light is exposed (Dickson et al., 2009). In order to test this model *in silico*, the genetically modified organisms will be tested under the same parameters. Ideally, they will behave as expected in stable conditions: the dark, anaerobic environment will create a natural situation for hydrogen production because no oxygen is present and the cell is not producing oxygen as a byproduct of photosynthesis. If the cell is without light but in an aerobic environment, the globin proteins on the membrane will keep the cytoplasm anaerobic. If the cell is photosynthesizing but in an anaerobic environment, the globin proteins should bind to the intracellular oxygen, and if the cell is producing oxygen in an oxygenated environment, then glbN needs to bind to both intracellular O_2 and O_2 diffusing through the membrane. The two trials with light will prove to be more difficult, as the cell will produce oxygen, so one would expect them to have a lower hydrogen yield than the cells in the dark.

Results and Discussion

Synechocystis PCC 6803 produces 0.025µM O_2 due to photosynthesis (Kihara et al., 2014). Thus, the first trial assumed that no environmental oxygen diffused through the cellular membrane, and it assumed the intracellular O_2 could be bound by glbN.

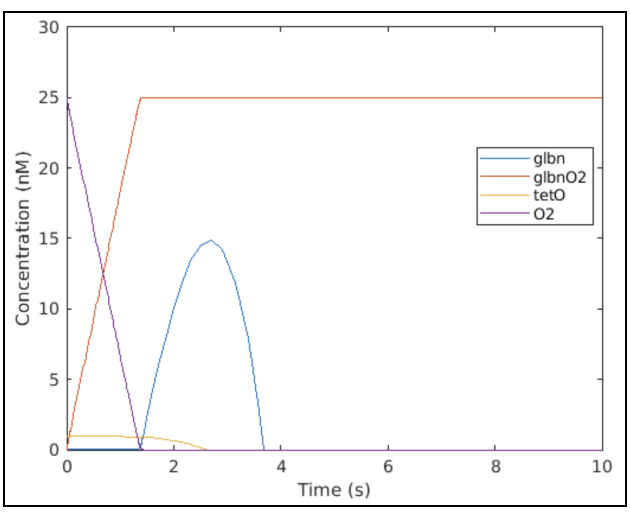

Figure 3: Computational model of a photosynthesizing cell in anaerobic conditions.

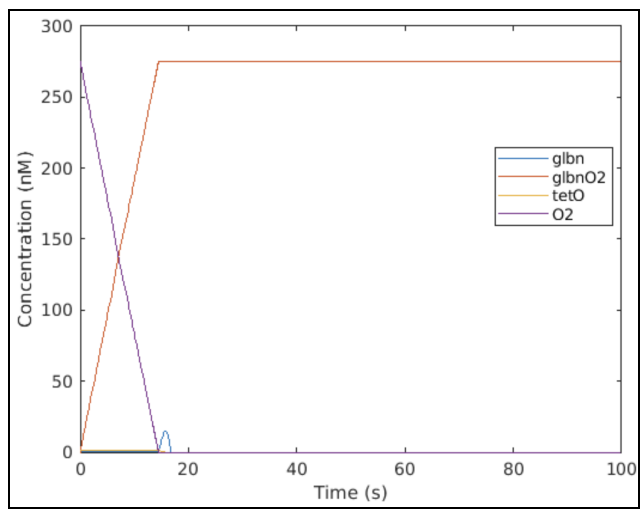

Figure 4: Computational model of a photosynthesizing cell in aerobic conditions.

As one can observe from Figure 3, the intracellular oxygen was immediately bound by globin. This is because tetO is constitutively expressed, and as soon as glbN is made, it binds to the available oxygen. This is favorable because O_2 inhibits hydrogen production whereas glbN simply places slightly more load on the cell, so it is beneficial to automatically produce globin immediately rather than wait to detect if O_2 is present or not. Once all of the oxygen has been bound by glbN to create glbN::O_2, glbN is still produced. However, once the upstream parts of

the genetic circuit detect that there is no oxygen left, tetR is produced and tetO is no longer able to create glbN, so the concentration of glbN decreases as it binds to O_2 that dissociated from glbN::O_2 and the protein degrades.

In water that comes into contact with air, the oxygen concentration is $2.25 * 10^{-4} M$. Thus, to simplify the system the extracellular concentration of oxygen will be combined with the intracellular concentration of oxygen to exemplify a cyanobacterial cell in light in an aerobic environment. This results in an oxygen concentration of $2.75 * 10^{-4} M$. As expected, Figure 4 shows that this trial yields a similar result to the first one, except for the fact that it takes slightly longer for the cell to produce enough glbN to bind to O_2.

For the cells placed in the dark, there will be no light energy to undergo photosynthesis, so the cell will not be producing oxygen. Thus, the only oxygen that could interfere with hydrogenase activity will be environmental O_2. This yields essentially the same result as the second trial, as depicted by Figure 5.

Finally, when *Synechocystis* is placed in a dark, anaerobic environment, it already produces H_2, so this circuit should not interfere with the cell. Figure 6 displays the fact that when no oxygen is present, glbN is repressed by the production of tetR. The glbN initially produced slowly decays away, and the cell is in its natural state.

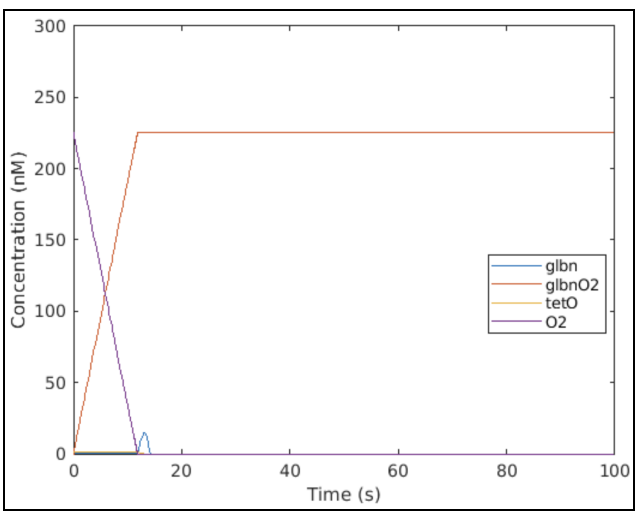

Figure 5: Computational model of a non-photosynthesizing cell in aerobic conditions.

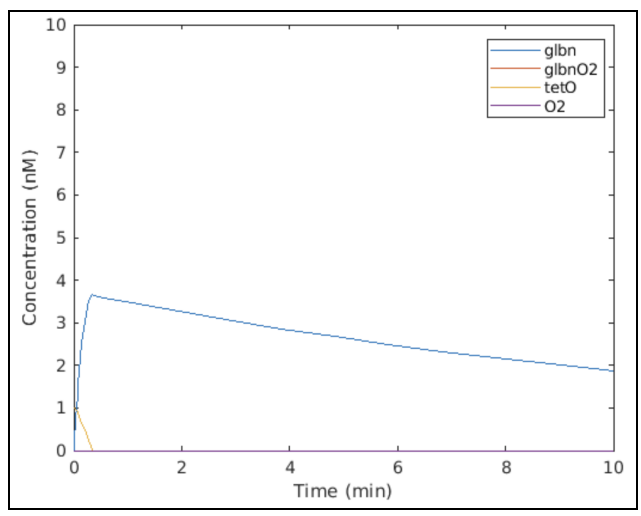

Figure 6: Computational model of a non-photosynthesizing cell in anaerobic conditions.

Looking at the results as a whole, this genetic circuit functions well under the assumptions made. Future steps in this field include addressing some of the assumptions. For instance, it should be investigated whether the glbN protein can prevent oxygen from entering the cell by residing on the membrane. If a complete anaerobic environment is created inside the cell, then there would exist a large chemical gradient for oxygen to diffuse into the cell. Furthermore, it was assumed that the oxygen inside the cell, produced as a byproduct of

photosynthesis, would be bound by glbN. However, that is not guaranteed. Since hydrogen production in *Synechocystis* occurs in the PQ pool, there might be no possible way to separate the O_2 produced in photosynthesis from the hydrogenase needed to produce H_2 because they operate in the same physical space. It would be of interest to make the space inside the thylakoid anaerobic so that hydrogenase could function while the cell was undergoing photosynthesis. This could be achieved with a chemical gradient in conjunction with the oxygen content of the cytoplasm. Another future paper would be to examine the off-target effects of the tetR protein. *Synechocystis* does have a homolog to tetR, and since tetR is being produced constantly in anaerobic environments, it could effect expression of other genes. That being said, the genetic circuit presented in this paper might be most applicable to instances without light. This is still advantageous because a mutant cell would ideally be able to operate in an oxygenated environment, whereas a normal cell could not.

Conclusion

The novel biological gene circuit presented here prevents O_2 from interfering with hydrogenase activity *in silico*. In environments where the cell is producing O_2 or O_2 is present in the environment, the circuit produces a globin protein that binds with high affinity to oxygen. When no oxygen is present, the circuit does not place a significantly larger load on the cell.

The future of cyanobacterial hydrogen production is promising; however, it is not close to becoming commercially viable yet. In addition to investigating the subtleties of this approach, hydrogen production in general needs to be optimized, including strategies surrounding the recycling of NAD(P)H through the NDH-1 complex and rapid biomass accumulation. Internal processes of *Synechocystis* PCC 6803 need to be more fully understood, and researchers need to document other, possibly more helpful species of cyanobacteria that possess the ability to produce hydrogen.

Appendix

Variable	Description	Reaction	Value	Source
K_{fnr-a}	Rate of O_2 deactivating *fnr*	O_2 + *fnr*–a → O_2 + *fnr*–d	4.09	Tolla et al., 2010; Unden et al., 2002
K_{fnr-d}	Rate of formation of active *fnr*	*fnr*–d + glutathione → *fnr*–a	0.245	Tran et al., 2001

$K_{fnr + fpromoter}$	Rate of *fnr* binding to its promoter	*fnr*_a + fpromoter → *fnr*::fpromoter	1.15	Tolla et al., 2010
$K_{fnr::fpromoter}$	Rate of *fnr* dissociating from its promoter	*fnr*::fpromoter → *fnr*_a + fpromoter	0.87	Tolla et al., 2010
K_{ftxn}	Rate of transcription of tetR	*fnr*::fpromoter→ *fnr*_a + fpromoter + tetR	11.5-23 min^{-1}	Dennis and Bremer, 2008
$K_{tetR + tetO}$	Rate of association of tetR and tetO	tetR + tetO → tetR::tetO	$5.58 \pm 2 \times 10^{9}$	Kamionka et al., 2004
$K_{tetR::tetO}$	Rate of dissociation of tetR and tetO	tetR::tetO → tetR + tetO	$1.79 \pm 2 \times 10^{7}$	Kamionka et al., 2004
K_{gtxn}	Rate of txn of glbN	tetO → tetO + glbN	19.4-38.7 min^{-1}	Dennis and Bremer, 2008
K_{pdeg}	Rate of degradation of protein	tetR → ∅ glbN → ∅	0.069 min^{-1}	Elowitz et al., 2000
$K_{glbN + O2}$	Rate of association between glbN and O_2	glbN + O_2 → glbN::O_2	390	Dou et al., 1998
$K_{glbN::O2}$	Rate of dissociation between glbN and O_2	glbN::O_2 → glbN + O_2	2.56×10^{-3}	Dou et al., 1998

Table 1: Values of parameters. It displays the parameter, its description, the reaction it takes place in, its value, and the reference.

$$\frac{d[fnr\text{–}a]}{dt} = -K_{fnr\text{–}a} * [fnr\text{–}a] * [O_2] + K_{fnr::fpromoter} * [fnr::fpromoter] + K_{ftxn} * [fnr::fpromoter] - K_{pdeg} * [fnr\text{–}a]$$

$$\frac{d[fnr\text{–}d]}{dt} = K_{fnr\text{–}a} * [fnr\text{–}a] * [O_2] - K_{fnr\text{–}d} * [fnr\text{–}d] - K_{pdeg} * [fnr\text{–}d]$$

$$\frac{d[fpromoter]}{dt} = K_{fnr::fpromoter} * [fnr::promoter] - K_{fnr + fpromoter} * [fnr] * [fpromoter] + K_{ftxn} * [fnr::fpromoter]$$

$$\frac{d[tetR]}{dt} = [fnr::fpromoter] * K_{ftxn} - K_{tetR+tetO} * [tetR] * [tetO] - K_{pdeg} * [tetR] + K_{tetR::tetO} * [tet::tetO]$$

$$\frac{d[tetO]}{dt} = K_{tetR::tetO} * [tetR::tetO] - K_{tetR+tetO} * [tetR] * [tetO]$$

$$\frac{d[tetR::tetO]}{dt} = K_{tetR+tetO} * [tetR] * [tetO] - K_{tetR::tetO}[tetR::tetO]$$

$$\frac{d[glbN]}{dt} = K_{gtxn} * [tetO] - K_{pdeg} * [glbN] + K_{glbN::O_2} * [glbN::O_2] - K_{glbN+O_2} * [glbN] *$$

$$\frac{d[glbN::O_2]}{dt} = K_{glbN+O_2} * [glbN] * [O_2] - K_{glbN::O_2} * [glbN::O_2]$$

$$\frac{dO_2}{dt} = K_{glbN::O_2} * [glbN::O_2] - K_{glbN+O_2} * [glbN] * [O_2]$$

Figure 2: Representing the system with ODEs. Each ordinary differential equation describes the aggregate change in concentration of a specific part.

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